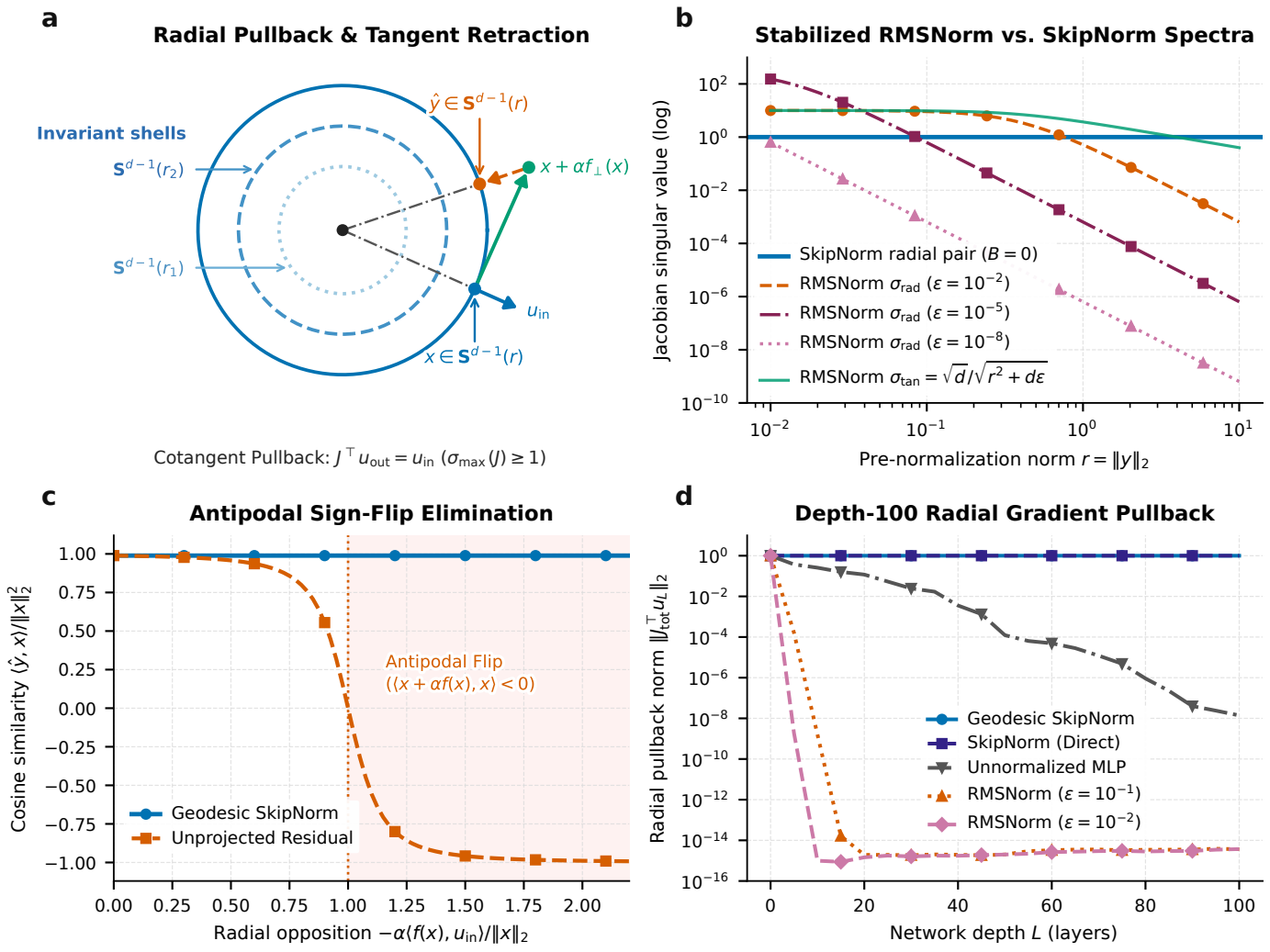

a
Radial Pullback & Tangent Retraction
Invariant shells
$\mathbf{S}^{d-1}(r_2)$
$\mathbf{S}^{d-1}(r_1)$
$\hat{y} \in \mathbf{S}^{d-1}(r)$
$x + \alpha f_\perp(x)$
u_{in}
$x \in \mathbf{S}^{d-1}(r)$
Cotangent Pullback: $J^\top u_{\text{out}} = u_{\text{in}}$ $(\sigma_{\max}(J) \geq 1)$
b
Stabilized RMSNorm vs. SkipNorm Spectra
Jacobian singular value (log)
SkipNorm radial pair ($B = 0$)
RMSNorm σ_{rad} ($\varepsilon = 10^{-2}$)
RMSNorm σ_{rad} ($\varepsilon = 10^{-5}$)
RMSNorm σ_{rad} ($\varepsilon = 10^{-8}$)
RMSNorm $\sigma_{\text{tan}} = \sqrt{d}/\sqrt{r^2 + d\varepsilon}$
Pre-normalization norm $r = \|y\|_2$
c
Antipodal Sign-Flip Elimination
Cosine similarity $\langle \hat{y}, x \rangle / \|x\|_2^2$
Antipodal Flip
$(\langle x + \alpha f(x), x \rangle < 0)$
Geodesic SkipNorm
Unprojected Residual
Radial opposition $-\alpha \langle f(x), u_{\text{in}} \rangle / \|x\|_2$
d
Depth-100 Radial Gradient Pullback
Radial pullback norm $\|J_{\text{tot}}^\top u_L\|_2$
Geodesic SkipNorm
SkipNorm (Direct)
Unnormalized MLP
RMSNorm ($\varepsilon = 10^{-1}$)
RMSNorm ($\varepsilon = 10^{-2}$)
Network depth L (layers)